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Substrate connector

Abstract

The present disclosure relates to a substrate connector comprising a plurality of radio frequency (RF) contacts for RF signal transmission; an insulation unit for supporting the RF contacts; a plurality of transmission contacts coupled to the insulation unit between a first RF contact and a second RF contact so the first RF contact of the RF contacts and the second RF contact of the RF contacts are spaced from each other in a first axial direction; a ground housing to which the insulation unit is coupled; a first ground contact which is coupled to the insulation unit; and a second ground contact which is coupled to the insulation unit, wherein the first RF contact includes a first RF inspection plane with which an inspection instrument is to come in contact, and the second RF contact includes a second inspection plane with which the inspection instrument is to come in contact.

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5718592	12/1997	Hosler, Sr.	439/83	H01R 12/716
7086866	12/2005	Folan	439/954	H01R 12/716
7198492	12/2006	Shichida	439/74	H01R 24/50
9065228	12/2014	Nishimura	N/A	H01R 13/6594
2006/0141811	12/2005	Shichida	439/63	H01R 24/50
2014/0287628	12/2013	Zhang et al.	N/A	N/A
2017/0265322	12/2016	Son	N/A	H01R 12/716
2018/0151967	12/2017	Chen	N/A	H01R 12/57
2019/0214772	12/2018	Kodama	N/A	H01R 13/6582
2019/0280439	12/2018	Ni	N/A	H01R 13/2421

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10069950	12/1997	JP	N/A
2018101559	12/2017	JP	N/A
20200123438	12/2019	JP	N/A
1020170015125	12/2016	KR	N/A
1020170129066	12/2016	KR	N/A
1020200041188	12/2019	KR	N/A
2020003731	12/2019	WO	N/A
2020123438	12/2019	WO	N/A

OTHER PUBLICATIONS

International Search Report for related International Application No. PCT/KR2021/009889; action dated Mar. 3, 2022; (5 pages). cited by applicant

Written Opinion for related International Application No. PCT/KR2021/009889; action dated Mar. 3, 2022; (8 pages). cited by applicant
Office Action for related Japanese Application No. 2023-513875; action dated Feb. 20, 2024; (3 pages). cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application is a National Stage of International Application No. PCT/KR2021/009889 filed on Jul. 29, 2021, which claims priority to and the benefit of Korean Utility Model Application No. 10-2020-0108965, filed on Aug. 28, 2020; and Korean Utility Model Application No. 10-2021-0095753, filed Jul. 21, 2021 the disclosures of which are incorporated herein by reference in their entirety.

FIELD

(2) The present disclosure relates to a substrate connector which is installed in an electronic device for electrical connection between substrates.

BACKGROUND

(3) A connector is provided in various electronic devices for electrical connection. For example, the connector is installed in an electronic device such as a mobile phone, a computer, a tablet computer and the like such that various parts installed in the electronic device can be electrically connected to each other.

(4) In general, among electronic devices, an RF connector and a board-to-board connector (hereinafter, referred to as a ‘substrate connector’) are provided inside wireless communication devices such as smart phones, tablet PCs and the like. The RF connector transmits a radio frequency (RF) signal. The substrate connector processes digital signals such as cameras.

(5) Such RF connectors and substrate connectors are mounted on a printed circuit board (PCB). Conventionally, since several substrate connectors and RF connectors are mounted together with a large number of components in a limited PCB space, there is a problem in that the PCB mounting area becomes large. Accordingly, in accordance with the trend of miniaturization of smartphones, there is a need for a technique for optimizing a small PCB mounting area by integrating an RF connector and a substrate connector.

(6) FIG. 1 is a schematic perspective view of a substrate connector according to the related art.

(7) Referring to FIG. 1, a substrate connector **100** according to the related art includes a first connector **110** and a second connector **120**.

(8) The first connector **110** is to be coupled to a first substrate (not illustrated). The first connector **110** may be electrically connected to the second connector **120** through a plurality of first contacts **111**.

(9) The second connector **120** is to be coupled to a second substrate (not illustrated). The second connector **120** may be electrically connected to the first connector **110** through a plurality of second contacts **121**.

(10) The substrate connector **100** according to the related art may electrically connect the first substrate and the second substrate to each other as the first contacts **111** and the second contacts

121 are connected to each other. In addition, when some contacts among the first contacts **111** and the second contacts **121** are used as RF contacts for RF signal transmission, the substrate connector **100** according to the related art may be implemented such that an RF signal is transmitted between the first substrate and the second substrate through the RF contact.

(11) Herein, the substrate connector **100** according to the related art has the following problems.

(12) First, in the substrate connector **100** according to the related art, when using contacts that are spaced apart by a relatively short distance among the contacts **111**, **121** as the RF contacts, there is a problem in that signal transmission is not performed smoothly between the RF contacts **111'**, **111''**, **121'**, **121''** due to RF signal interference.

(13) Second, since the substrate connector **100** according to the related art has an RF signal shielding part **112** in the outermost part of the connector, radiation to the outside of the RF signal can be shielded, but the shielding between the RF signals is not performed.

(14) Third, in the substrate connector **100** according to the related art, the RF contacts **111'**, **111''**, **121'**, **121''** include mounting parts **111a'**, **111a''**, **121a'**, **121a''** mounted on the substrate, respectively, and the mounting parts **111a'**, **111a''**, **121a'**, **121a''** are disposed to be exposed to the outside. Accordingly, the substrate connector **100** according to the related art has a problem in that the shielding for the mounting parts **111a'**, **111a''**, **121a'**, **121a''** is not performed.

SUMMARY

(15) The present disclosure has been devised to solve the above-described problems, and an object of the present disclosure to provide a substrate connector which is capable of reducing the possibility of generating RF signal interference between RF contacts.

(16) In order to solve the above problems, the present disclosure may include the following configurations.

(17) The substrate connector according to the present disclosure may include a plurality of radio frequency (RF) contacts for RF signal transmission; an insulation unit for supporting the RF contacts; a plurality of transmission contacts coupled to the insulation unit between a first RF contact and a second RF contact such that the first RF contact of the RF contacts and the second RF contact of the RF contacts are spaced apart from each other in a first axial direction; a ground housing to which the insulation unit is coupled; a first ground contact which is coupled to the insulation unit and shields the gap between the first RF contact and the transmission contacts on the basis of the first axial direction; and a second ground contact which is coupled to the insulation unit and shields the gap between the second RF contact and the transmission contacts on the basis of the first axial direction. The first RF contact may include a first RF inspection plane with which an inspection instrument is to come in contact. The second RF contact may include a second inspection plane with which the inspection instrument is to come in contact. The first RF inspection plane and the second RF inspection plane may be arranged on a plane having the same height.

(18) According to the present disclosure, the following effects can be achieved.

(19) The present disclosure can implement a shielding function of signals, electromagnetic waves and the like for RF contacts by using the grounding housing. Accordingly, the present disclosure can prevent electromagnetic waves generated from RF contacts from interfering with signals of circuit components located in the vicinity of the electronic device, and can prevent electromagnetic waves generated from circuit components located in the vicinity of the electronic device from interfering with signals transmitted by the RF contacts. Therefore, the present disclosure can contribute to improving EMI (Electro Magnetic Interference) shielding performance and EMC (Electro Magnetic Compatibility) performance by using the grounding housing.

(20) The present disclosure may be implemented such that all of the RF contacts including the portion mounted on a substrate are located inside the ground housing. Accordingly, the present disclosure can implement complete shielding by reinforcing the shielding function for RF contacts by using the grounding housing.

(21) The present disclosure is implemented such that an area for contacting an inspection

instrument can be secured, thereby improving the ease and accuracy of contact with the inspection instrument. Accordingly, the present disclosure can improve the accuracy of the inspection results by using an inspection instrument.

(22) In the present disclosure, since the surface to which the inspection instrument is in contact is formed as a flat surface, it is possible to prevent slip or the like from occurring in the process of contacting the inspection instrument. Therefore, the present disclosure can improve the contact reliability for the inspection instrument.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic perspective view of a substrate connector according to the related art.
- (2) FIG. 2 is a schematic perspective view of a receptacle connector and a plug connector in the substrate connector according to the present disclosure.
- (3) FIG. 3 is a schematic perspective view of the substrate connector according to the first example.
- (4) FIG. 4 is a schematic exploded perspective view of the substrate connector according to the first example.
- (5) FIG. 5 is a schematic plan view of the substrate connector according to the first example.
- (6) FIG. 6 is a schematic plan view of contacts in the substrate connector according to the first example.
- (7) FIGS. 7 and 8 are schematic plan views showing enlarged inspection planes of contacts in the substrate connector according to the first example.
- (8) FIG. 9 is a schematic plan view of contacts in the substrate connector according to the first example.
- (9) FIG. 10 is a schematic cross-sectional view taken along line I-I of FIG. 6.
- (10) FIG. 11 is a schematic cross-sectional view taken along line II-II of FIG. 6.
- (11) FIG. 12 is a schematic cross-sectional view taken along line of FIG. 9.
- (12) FIG. 13 is a schematic cross-sectional view taken along the line VI-VI of FIG. 9.
- (13) FIG. 14 is a schematic plan view showing a modified exemplary embodiment of the arrangement of contacts in the substrate connector according to the first example.
- (14) FIG. 15 is a conceptual plan view for explaining a ground loop in the substrate connector according to the first example.
- (15) FIG. 16 is a schematic side cross-sectional view showing a state in which the substrate connector according to the first example and the substrate connector according to the second example are spaced apart from each other for coupling based on the line A-A of FIG. 15.
- (16) FIG. 17 is a schematic perspective view of the substrate connector according to the second example.
- (17) FIG. 18 is a schematic exploded perspective view of the substrate connector according to the second example.
- (18) FIG. 19 is a schematic plan view of the substrate connector according to the second example.
- (19) FIG. 20 is a schematic plan view of contacts in the substrate connector according to the second example.
- (20) FIG. 21 is a conceptual plan view for explaining a ground loop in the substrate connector according to the second example.

DETAILED DESCRIPTION

(21) Hereinafter, the exemplary embodiments of a substrate connector according to the present disclosure will be described in detail with reference to the accompanying drawings. In FIG. 14, the positions of portions corresponding to the inspection plane in the contacts are indicated by dotted squares. FIG. 16 shows the substrate connector according to the second example in a state of being

reversed in the directions illustrated in FIGS. 2 and 17 and coupled to the substrate connector according to the first example. FIG. 16 is a schematic side cross-sectional view of the substrate connector according to a second example taken along line B-B of FIG. 19.

(22) Referring to FIG. 2, the substrate connector **1** according to the present disclosure may be installed in an electronic device (not illustrated) such as a mobile phone, a computer, a tablet computer or the like. The substrate connector **1** according to the present disclosure may be used to electrically connect a plurality of substrates (not illustrated). The substrates may be printed circuit boards (PCBs). For example, when the first substrate and the second substrate are electrically connected, a receptacle connector mounted on the first substrate and a plug connector mounted on the second substrate may be connected to each other. Accordingly, the first substrate and the second substrate may be electrically connected to each other through the receptacle connector and the plug connector. A plug connector mounted on the first substrate and a receptacle connector mounted on the second substrate may be connected to each other.

(23) The substrate connector **1** according to the present disclosure may be implemented as the receptacle connector. The substrate connector **1** according to the present disclosure may be implemented as the plug connector. The substrate connector **1** according to the present disclosure may be implemented by including both the receptacle connector and the plug connector.

Hereinafter, an exemplary embodiment in which the substrate connector **1** according to the present disclosure is implemented as the receptacle connector is defined as the substrate connector **200** according to the first example, and an exemplary embodiment in which the substrate connector **1** according to the present disclosure is implemented as the plug connector is defined as the substrate connector **300** according to the second example to be described in detail with reference to the accompanying drawings. In addition, an exemplary embodiment in which the substrate connector **200** according to the first example is mounted on the first substrate and the substrate connector **300** according to the second example is mounted on the second substrate will be described as a reference. From this, it will be apparent to those skilled in the art to derive an exemplary embodiment in which the substrate connector **1** according to the present disclosure includes both the receptacle connector and the plug connector.

Substrate Connector 200 According to the First Example

(24) Referring to FIGS. 2 to 4, the substrate connector **200** according to the first example may include a plurality of RF contacts **210**, a plurality of transmission contacts **220**, a ground housing **230** and an insulation unit **240**.

(25) The RF contacts **210** are for transmitting a radio frequency (RF) signal. The RF contacts **210** may transmit a very high-frequency RF signal. The RF contacts **210** may be supported by the insulation unit **240**. The RF contacts **210** may be coupled to the insulation unit **240** through an assembly process. The RF contacts **210** may be integrally molded with the insulation unit **240** through injection molding.

(26) The RF contacts **210** may be disposed to be spaced apart from each other. The RF contacts **210** may be electrically connected to the first substrate by being mounted on the first substrate. The RF contacts **210** may be electrically connected to the second substrate on which the counterpart connector is mounted by being connected to the RF contacts of the counterpart connector. Accordingly, the first substrate and the second substrate may be electrically connected. When the substrate connector **200** according to the first example is a receptacle connector, the counterpart connector may be a plug connector. When the substrate connector **200** according to the first example is a plug connector, the counterpart connector may be a receptacle connector.

(27) A first RF contact **211** of the RF contacts **210** and a second RF contact **212** of the RF contacts **210** may be spaced apart from each other in a first axial direction (X-axis direction). The first RF contact **211** and the second RF contact **212** may be supported by the insulation unit **240** at positions spaced apart from each other in the first axial direction (X-axis direction).

(28) The first RF contact **211** may include a first RF mounting member **2111**. The first RF

mounting member **2111** may be mounted on the first substrate. Accordingly, the first RF contact **211** may be electrically connected to the first substrate through the first RF mounting member **2111**. The first RF contact **211** may be formed of a material having electrical conductivity. For example, the first RF contact **211** may be formed of a metal. The first RF contact **211** may be connected to any one of the RF contacts of the counterpart connector.

(29) The second RF contact **212** may include a second RF mounting member **2121**. The second RF mounting member **2121** may be mounted on the first substrate. Accordingly, the second RF contact **212** may be electrically connected to the first substrate through the second RF mounting member **2121**. The second RF contact **212** may be formed of a material having electrical conductivity. For example, the second RF contact **212** may be formed of a metal. The second RF contact **212** may be connected to any one of the RF contacts of the counterpart connector.

(30) Referring to FIGS. 2 to 5, the transmission contacts **220** are coupled to the insulation unit **240**. The transmission contacts **220** may be in charge of transmitting a signal, data, power and the like. The transmission contacts **220** may be coupled to the insulation unit **240** through an assembly process. The transmission contacts **220** may be integrally molded with the insulation unit **240** through injection molding.

(31) The transmission contacts **220** may be disposed between the first RF contact **211** and the second RF contact **212** on the basis of the first axial direction (X-axis direction). Accordingly, the transmission contact **220** may be placed in a space spaced apart from the first RF contact **211** and the second RF contact **212** to reduce RF signal interference between the first RF contact **211** and the second RF contact **212**. Therefore, the substrate connector **200** according to the first example may not only reduce RF signal interference by increasing the distance between the first RF contact **211** and the second RF contact **212** spaced apart from each other, but also it is possible to improve the space utilization of the insulator **240** by disposing the transmission contacts **220** in a space which is spaced apart for this purpose.

(32) The transmission contacts **220** may be disposed to be spaced apart from each other. The transmission contacts **220** may be electrically connected to the first substrate by being mounted on the first substrate. In this case, the transmission mounting members **2211**, **2221** of each of the transmission contacts **220** may be mounted on the first substrate. The transmission contacts **220** may be formed of a material having electrical conductivity. For example, the transmission contacts **220** may be formed of a metal. The transmission contacts **220** may be electrically connected to the second substrate on which the counterpart connector is mounted by being connected to the transmission contacts of the counterpart connector. Accordingly, the first substrate and the second substrate may be electrically connected.

(33) The first transmission contacts **221** of the transmission contacts **220** and the second transmission contacts **222** of the transmission contacts **220** may be disposed to be spaced apart from each other in the second axial direction (Y-axis direction). The second axial direction (Y-axis direction) is an axial direction perpendicular to the first axial direction (X-axis direction). The first transmission contacts **221** may be disposed to be spaced apart from each other in the first axial direction (X-axis direction). The second transmission contacts **222** may be disposed to be spaced apart from each other in the first axial direction (X-axis direction).

(34) Referring to FIGS. 2 to 5, the ground housing **230** is coupled to the insulation unit **240**. The ground housing **230** may be grounded by being mounted on the first substrate. Accordingly, the ground housing **230** may implement a shielding function for signals, electromagnetic waves and the like for the RF contacts **210**. In this case, the ground housing **230** may prevent electromagnetic waves generated from the RF contacts **210** from interfering with signals of circuit components located in the vicinity of the electronic device, and may prevent electromagnetic waves generated from circuit components located in the vicinity of the electronic device from interfering with the RF signals transmitted by the RF contacts **210**. Accordingly, the substrate connector **200** according to the first example may contribute to improving the EMI (Electro Magnetic Interference) shielding

performance and EMC (Electro Magnetic Compatibility) performance by using the ground housing **230**. The ground housing **230** may be formed of a material having electrical conductivity. For example, the ground housing **230** may be formed of a metal.

(35) The ground housing **230** may be disposed to surround the side of the inner space **230a**. A portion of the insulation unit **240** may be positioned in the inner space **230a**. All of the first RF contact **211**, the second RF contact **212** and the transmission contact **220** may be located in the inner space **230a**. In this case, all of the first RF mounting member **2111**, the second RF mounting member **2121** and the transmission mounting members **2211**, **2221** may also be located in the inner space **230a**. Accordingly, the ground housing **230** implements a shielding wall for all of the first RF contact **211** and the second RF contact **212**, and thus, it is possible to realize complete shielding by strengthening the shielding function for the first RF contact **211** and the second RF contact **212**. The counterpart connector may be inserted into the inner space **230a**.

(36) The ground housing **230** may be disposed to surround all sides with respect to the inner space **230a**. The inner space **230a** may be disposed inside the ground housing **230**. When the ground housing **230** is formed in a rectangular ring shape as a whole, the inner space **230a** may be formed in a rectangular parallelepiped shape. In this case, the ground housing **230** may be disposed to surround four sides with respect to the inner space **230a**.

(37) The ground housing **230** may be integrally formed without a seam. The ground housing **230** may be integrally formed without a seam by the metal injection method such as the metal die casting method or the metal injection molding (MIM) method. The ground housing **230** may be integrally formed without a seam by CNC (Computer Numerical Control) machining, MCT (Machining Center Tool) machining or the like.

(38) Referring to FIGS. 2 to 5, the insulation unit **240** supports the RF contacts **210**. The RF contacts **210** and the transmission contacts **220** may be coupled to the insulation unit **240**. The insulation unit **240** may be formed of an insulating material. The insulation unit **240** may be coupled to the ground housing **230** such that the RF contacts **210** are positioned in the inner space **230a**.

(39) Referring to FIGS. 2 to 6, the substrate connector **200** according to the first example may include a first ground contact **250**.

(40) The first ground contact **250** is coupled to the insulation unit **240**. The first ground contact **250** may be grounded by being mounted on the first substrate. The first ground contact **250** may be coupled to the insulation unit **240** through an assembly process. The first ground contact **250** may be integrally molded with the insulation unit **240** through injection molding.

(41) The first ground contact **250** may implement a shielding function for the first RF contact **211** together with the ground housing **230**. In this case, the first ground contact **250** may be disposed between the first RF contact **211** and the transmission contacts **220** on the basis of the first axial direction (X-axis direction). The first ground contact **250** may be formed of a material having electrical conductivity. For example, the first ground contact **250** may be formed of a metal. When the counterpart connector is inserted into the inner space **230a**, the first ground contact **250** may be connected to a ground contact of the counterpart connector.

(42) Referring to FIGS. 2 to 5, the substrate connector **200** according to the first example may include a second ground contact **260**.

(43) The second ground contact **260** is coupled to the insulation unit **240**. The second ground contact **260** may be grounded by being mounted on the first substrate. The second ground contact **260** may be coupled to the insulation unit **240** through an assembly process. The second ground contact **260** may be integrally molded with the insulation unit **240** through injection molding.

(44) The second ground contact **260** may implement a shielding function for the second RF contact **212** together with the ground housing **230**. The second ground contact **260** may be disposed between the transmission contacts **220** and the second RF contact **212** on the basis of the first axial direction (X-axis direction). The second ground contact **260** may be formed of a material having

electrical conductivity. For example, the second ground contact **260** may be formed of a metal. When the counterpart connector is inserted into the inner space **230a**, the second ground contact **260** may be connected to a ground contact of the counterpart connector.

(45) Herein, the substrate connector **200** according to the first example may be inspected by an inspection instrument (not illustrated) in order to check whether performance is abnormal in a state where it is mounted on the first substrate. The inspection instrument may perform the inspection on the substrate connector **200** according to the first example in a state where the probes are brought into contact with at least one of the contacts **210**, **220**, **250**, **260**. For example, the inspection instrument may be a probe card. As described above, the substrate connector **200** according to the first example may be implemented as follows such that the inspection through the contact of the inspection instrument can be smoothly performed.

(46) First, the first RF contact **211** may be implemented as follows.

(47) Referring to FIGS. **2** to **10**, the first RF contact **211** may include a first RF inspection plane **2112**.

(48) The first RF inspection plane **2112** is for contacting the inspection instrument. The inspection instrument may perform an inspection on the first RF contact **211** in a state in contact with the first RF inspection plane **2112**. The first RF inspection plane **2112** may be formed as a flat surface. Accordingly, since the substrate connector **200** according to the first example may secure an area for the inspection instrument to contact the first RF contact **211** by using the first RF inspection plane **2112**, it is possible to improve the ease and accuracy of the contact between the inspection instrument and the first RF contact **211**. Therefore, the substrate connector **200** according to the first example may improve the accuracy of the inspection result for the first RF contact **211** by using the first RF inspection plane **2112**. Further, in the case of the comparative example in which the surface of the first RF contact **211** for contacting the inspection instrument is formed as a curved surface, slip or the like occurs while the inspection instrument is in contact with the curved surface, and thus, the contact reliability between the inspection instrument and the first RF contact **211** may be reduced. In contrast, in the substrate connector **200** according to the first example, since the first RF inspection plane **2112** is formed as a flat surface, it is implemented to prevent slip or the like from occurring in the process of the inspection instrument being in contact with the first RF inspection plane **2112**. Accordingly, the substrate connector **1** according to the first example may improve the contact reliability between the inspection instrument and the first RF contact **211**.

(49) When the width H (illustrated in FIGS. **7** and **8**) based on the first axial direction (X-axis direction) of the first RF inspection plane **2112** is 1, it may be formed as a plane with the length L (illustrated in FIGS. **7** and **8**) of 0.5 or more based on the second axial direction (Y-axis direction). If the ratio of the length L to the width H of the first RF inspection plane **2112** is less than 0.5, the length L of the first RF inspection plane **2112** is too short, and thus, it is difficult to make accurate contact between the inspection instrument and the first RF inspection **2112**. In consideration of this, in the substrate connector **200** according to the first example, the first RF inspection plane **2112** is formed in a plane in which the ratio of the length L to the width H is 0.5 or more, and thus, it is possible to improve the ease and accuracy of the contact between the inspection instrument and the first RF contact **211**.

(50) When the width H (illustrated in FIGS. **7** and **8**) based on the first axial direction (X-axis direction) of the first RF inspection plane **2112** is 1, it may be formed as a plane with the length L (illustrated in FIGS. **7** and **8**) of 1.5 or less based on the second axial direction (Y-axis direction). If the ratio of the length L to the width H of the first RF inspection plane **2112** is more than 1.5, the length of the first RF contact **211** is too long due to the first RF inspection plane **2112**, and thus, it is difficult to realize miniaturization. In consideration of this, in the substrate connector **200** according to the first example, the first RF inspection plane **2112** is formed in a plane in which the ratio of the length L to the width H is 1.5 or less, thereby realizing miniaturization.

(51) When the width H (illustrated in FIGS. **7** and **8**) based on the first axial direction (X-axis

direction) of the first RF inspection plane **2112** is 1, it may be formed as a plane with the length L (illustrated in FIGS. 7 and 8) of 0.5 or more to 1.5 or less based on the second axial direction (Y-axis direction). That is, the first RF inspection plane **2112** may be formed in a plane in which the ratio of the length L to the width H is 0.5 or more to 1.5 or less. Accordingly, the substrate connector **200** according to the first example may not only secure an area for the inspection instrument to contact the first RF contact **211** by using the first RF inspection plane **2112**, but also it is possible to realize miniaturization by reducing the overall size.

(52) The first RF contact **211** may include a first RF connecting member **2113**.

(53) The first RF connecting member **2113** is for connecting to the RF contact of the counterpart connector. As the first RF connecting member **2113** is connected to the RF contact of the counterpart connector, the first RF contact **211** may be electrically connected to the RF contact of the counterpart connector. The first RF connecting member **2113** may be connected to the first RF mounting member **2111**. The first RF connecting member **2113** may be directly coupled to the first RF mounting member **2111**. The first RF connecting member **2113** is coupled to another member of the first RF connecting member **2113**, and may be connected to the first RF mounting member **2111** through the member. The first RF connecting member **2113** may be implemented in a form where a plate material disposed in a vertical direction and a plate material disposed in a horizontal direction are combined through bending processing for the plate material.

(54) When the first RF contact **211** includes the first RF connecting member **2113** and the first RF mounting member **2111**, the first RF inspection plane **2112** may be formed on the first RF connecting member **2113** or the first RF mounting member **2111**. As illustrated in FIG. 9, when the first RF inspection plane **2112** is formed on the first RF connecting member **2113**, the first RF inspection plane **2112** may be formed on a surface of the first RF connecting member **2113** facing the inner space **230a**. In this case, the first RF connecting member **2113** may have a connection function for the RF contact of the counterpart connector and a contact function for the inspection instrument. Although not illustrated, when the first RF inspection plane **2112** is formed on the first RF mounting member **2111**, the first RF inspection plane **2112** may be formed on a surface of the first RF mounting member **2111** facing the inner space **230a**. In this case, the first RF mounting member **2111** may have a mounting function for the substrate and a contact function for the inspection instrument.

(55) The first RF contact **211** may include a first RF inspection member **2114**.

(56) The first RF inspection member **2114** is for contacting the inspection instrument. In this case, as illustrated in FIGS. 5 and 6, the first RF inspection plane **2112** may be formed on the first RF inspection member **2114**. The first RF inspection plane **2112** may be formed on a surface of the first RF inspection member **2114** facing the inner space **230a**. The first RF inspection member **2114** may be coupled to the first RF connecting member **2113** or the first RF mounting member **2111**. As illustrated in FIG. 6, the first RF inspection member **2114** may be coupled to the first RF connecting member **2113** to protrude from the first RF connecting member **2113**. In this case, the first RF inspection member **2114** may have a support function supported by the insulation unit **240** and a contact function for the inspection instrument. The first RF inspection member **2114** may be coupled to each of the first RF connecting member **2113** and the first RF mounting member **2111**. In this case, the first RF inspection member **2114** may have a connection function for connecting the first RF connecting member **2113** and the first RF mounting member **2111** and a contact function for the inspection instrument. Even when the first RF inspection member **2114** is provided, the first RF inspection plane **2112** may be formed on the first RF mounting member **2111** or the first RF connecting member **2113**. In this case, the first RF inspection member **2114** may have a support function for the insulation unit **240** or a connection function for connecting the first RF connecting member **2113** and the first RF mounting member **2111**.

(57) Meanwhile, when the first RF inspection plane **2112** is formed in a portion of the first RF contact **211** that is covered by the insulation unit **240**, an inspection connection window (not

illustrated) may be formed in the insulation unit **240**. Accordingly, the first RF inspection plane **2112** may be disposed to be exposed to the inner space **230a** without being covered by the insulation unit **240** through the inspection connection window.

(58) Next, the second RF contact **212** may be implemented as follows.

(59) Referring to FIGS. **2** to **14**, the second RF contact **212** may include a second RF inspection plane **2122**. The second RF inspection plane **2122** is for contacting the inspection instrument. Since the second RF inspection plane **2122** is implemented to substantially coincide with the first RF inspection plane **2112** of the first RF contact **211** as described above, the detailed description thereof will be omitted.

(60) The second RF contact **212** may include a second RF connecting member **2123**. The second RF connecting member **2123** is for connecting to the RF contact of the counterpart connector. Since the second RF connecting member **2123** is implemented to substantially coincide with the first RF connecting member **2113** of the first RF contact **211** as described above, the detailed description thereof will be omitted. The second RF inspection plane **2122** may be formed on the second RF connecting member **2123** or the second RF mounting member **2121**.

(61) The second RF contact **212** may include a second RF inspection member **2124**. The second RF inspection member **2124** is for contacting the inspection instrument. Since the second RF inspection member **2124** is implemented to substantially coincide with the first RF inspection member **2114** of the first RF contact **211** as described above, the detailed description thereof will be omitted. The second RF inspection plane **2122** may be formed on the second RF inspection member **2124**.

(62) As illustrated in FIGS. **10** to **13**, the second RF inspection plane **2122** and the first RF inspection plane **2112** may be disposed on a plane having the same height. Accordingly, since the height at which the inspection instrument is raised and lowered to contact each of the second RF inspection plane **2122** and the first RF inspection plane **2112** may be implemented to be the same, the substrate connector **1** according to the first example may further improve the ease and accuracy of the inspection operation of the first RF contact **211** and the second RF contact **212**. Further, in the substrate connector **1** according to the first example, since it is possible for the inspection instrument to contact the second RF inspection plane **2122** and the first RF inspection plane **2112** at the same time, it may contribute to shorten the time taken to perform the inspection operation of the first RF contact **211** and the second RF contact **212**.

(63) Meanwhile, the portion of the second RF contact **212** on which the second RF inspection plane **2122** is formed and the portion of the first RF contact **211** on which the first RF inspection plane **2112** is formed may have the same function. For example, as illustrated in FIG. **6**, when the second RF inspection plane **2122** is formed on the second RF inspection member **2124**, the first RF inspection plane **2112** may be formed on the first RF inspection member **2114**. For example, when the second RF inspection plane **2122** is formed on the second RF connecting member **2123** as illustrated in FIG. **9**, the first RF inspection plane **2112** may be formed on the first RF connecting member **2113**. For example, when the second RF inspection plane **2122** is formed on the second RF mounting member **2121**, the first RF inspection plane **2112** may be formed on the first RF mounting member **2111**.

(64) Next, each of the first transmission contacts **221** may be implemented as follows.

(65) Referring to FIGS. **2** to **14**, the first transmission contact **221** may include a first transmission mounting member **2211**.

(66) The first transmission mounting member **2211** is for mounting on the first substrate. The first transmission contact **221** may be electrically connected to the first substrate through the first transmission mounting member **2211**. The first transmission mounting member **2211** may be formed in a plate shape arranged in a horizontal direction.

(67) The first transmission contact **221** may include a first transmission inspection plane **2212**.

(68) The first transmission inspection plane **2212** is for contacting the inspection instrument. The

inspection instrument may perform the inspection on the first transmission contact **221** while being in contact with the first transmission inspection plane **2212**. The first transmission inspection plane **2212** may be formed as a flat surface. Accordingly, since the substrate connector **200** according to the first example may secure an area for the inspection instrument to contact the first transmission contact **221** by using the first transmission inspection plane **2212**, it is possible to improve the ease and accuracy of the contact between the inspection instrument and the first transmission contact **221**. Accordingly, in the substrate connector **200** according to the first example, the accuracy of the inspection result for the first transmission contact **221** may be improved by using the first transmission inspection plane **2212**. Further, in comparison with the comparative example in which the surface of the first transmission contact **221** for contacting the inspection instrument is formed as a curved surface, since the first transmission inspection plane **2212** is formed as a plane in the substrate connector **200** according to the first example, it is implemented to prevent slip or the like from occurring in the process of contacting the inspection instrument with the first transmission inspection plane **2212**. Accordingly, the substrate connector **1** according to the first example may improve the contact reliability between the inspection instrument and the first transmission contact **221**.

(69) When the width H (illustrated in FIGS. 7 and 8) of the first transmission inspection plane **2212** on the basis of the first axial direction (X-axis direction) is 1, it may be formed as a plane with the length L (illustrated in FIGS. 7 and 8) of 0.5 or more based on the second axial direction (Y-axis direction). When the width H (illustrated in FIGS. 7 and 8) of the first transmission inspection plane **2212** on the basis of the first axial direction (X-axis direction) is 1, it may be formed as a plane with the length L (illustrated in FIGS. 7 and 8) of 1.5 or less based on the second axial direction (Y-axis direction). When the width H (illustrated in FIGS. 7 and 8) of the first transmission inspection plane **2212** on the basis of the first axial direction (X-axis direction) is 1, it may be formed as a plane with the ratio of the length L to the height H of 0.5 or more to 1.5 or less based on the second axial direction (Y-axis direction). That is, the first transmission inspection plane **2212** may be formed as a plane in which the ratio of the length L to the width H is 0.5 or more to 1.5 or less. Accordingly, the substrate connector **200** according to the first example may not only secure an area for the inspection instrument to contact the first transmission contact **221** by using the first transmission inspection plane **2212**, but also it is possible to realize miniaturization by reducing the overall size.

(70) The first transmission contact **221** may include a first transmission connecting member **2213**.

(71) The first transmission connecting member **2213** is for connecting to the transmission contact of the counterpart connector. As the first transmission connecting member **2213** is connected to the transmission contact of the counterpart connector, the first transmission contact **221** may be electrically connected to the transmission contact of the counterpart connector. The first transmission connecting member **2213** may be connected to the first transmission mounting member **2211**. The first transmission connecting member **2213** may be directly coupled to the first transmission mounting member **2211**. The first transmission connecting member **2213** may be coupled to another member of the first transmission connecting member **2213** and may be connected to the first transmission mounting member **2211** through the member. The first transmission connecting member **2213** may be implemented in a form where a plate material disposed in a vertical direction and a plate material disposed in a horizontal direction are combined through bending processing for the plate material.

(72) The first transmission contact **221** may include a first transmission inspection member **2214**.

(73) The first transmission inspection member **2214** is for contacting the inspection instrument. In this case, as illustrated in FIGS. 5 and 6, the first transmission inspection plane **2212** may be formed on the first transmission inspection member **2214**. The first transmission inspection plane **2212** may be formed on a surface of the first transmission inspection member **2214** facing the inner space **230a**. The first transmission inspection member **2214** may be coupled to each of the first

transmission connecting member **2213** and the first transmission mounting member **2211**. In this case, the first transmission inspection member **2214** may have a connection function for connecting the first transmission connecting member **2213** and the first transmission mounting member **2211** and a contact function for the inspection instrument. The first transmission inspection member **2214** may be coupled to the first transmission connecting member **2213** or the first transmission mounting member **2211**. In this case, the first transmission inspection member **2214** may have a support function supported by the insulation unit **240** and a contact function for the inspection instrument.

(74) Although not illustrated, even when the first transmission inspection member **2214** is provided, the first transmission inspection plane **2212** may be formed on the first transmission mounting member **2211** or the first transmission connecting member **2213**. When the first transmission inspection plane **2212** is formed on the first transmission mounting member **2211**, the first transmission mounting member **2211** may have a mounting function to the substrate and a contact function to the inspection instrument. When the first transmission inspection plane **2212** is formed on the first transmission connecting member **2213**, the first transmission connecting member **2213** has a connection function for a connection contact of the counterpart connector and a contact function for the inspection instrument. When the first transmission inspection plane **2212** is formed on the first transmission mounting member **2211** or the first transmission connecting member **2213**, the first transmission inspection member **2214** may have a support function for the insulating part **240** or a connection function for connecting the first transmission connecting member **2213** and the first transmission mounting member **2211**.

(75) Meanwhile, when the first transmission inspection plane **2212** is formed in a portion of the first transmission contact **221** that is covered by the insulation unit **240**, an inspection connection window (not illustrated) may be formed on the insulation unit **240**. Accordingly, the first transmission inspection plane **2212** may be disposed to be exposed to the inner space **230a** without being covered by the insulation unit **240** through the inspection connection window.

(76) Next, each of the second transmission contacts **222** may be implemented as follows.

(77) Referring to FIGS. **2** to **14**, the second transmission contact **222** may include a second transmission mounting member **2221**. The second transmission mounting member **2221** is for mounting on the first substrate. Since the second transmission mounting member **2221** is implemented to substantially coincide with the first transmission mounting member **2211** of the first transmission contact **221** as described above, the detailed description thereof will be omitted.

(78) The second transmission contact **222** may include a second transmission inspection plane **2222**. The second transmission inspection plane **2222** is for contacting the inspection instrument. Since the second transmission inspection plane **2222** is implemented to substantially coincide with the first transmission inspection plane **2212** of the first transmission contact **221** as described above, the detailed description thereof will be omitted.

(79) The second transmission contact **222** may include a second transmission connecting member **2223**. The second transmission connecting member **2223** is for connecting to the transmission contact of the counterpart connector. Since the second transmission connecting member **2223** is implemented to substantially coincide with the first transmission connecting member **2213** of the first transmission contact **221** as described above, the detailed description thereof will be omitted.

(80) The second transmission contact **222** may include a second transmission inspection member **2224**. The second transmission inspection member **2224** is for contacting the inspection instrument. Since the second transmission inspection member **2224** is implemented to substantially coincide with the first transmission inspection member **2214** of the first transmission contact **221** as described above, the detailed description thereof will be omitted. The second transmission inspection plane **2222** may be formed on any one of the second transmission inspection member **2224**, the second transmission connecting member **2223** and the second transmission mounting member **2221**.

(81) As illustrated in FIGS. **10** to **13**, the second transmission inspection plane **2222** and the first transmission inspection plane **2212** may be disposed on a plane having the same height. Accordingly, since the height at which the inspection instrument is raised and lowered to come into contact with each of the second transmission inspection plane **2222** and the first transmission inspection plane **2212** may be implemented to be the same, the substrate connector **1** according to the first example may further improve the ease and accuracy of the inspection operation of the first transfer contact **221** and the second transfer contact **222**. Further, in the substrate connector **1** according to the first example, since the inspection instrument may contact the second transmission inspection plane **2222** and the first transmission inspection plane **2212** at the same time, it may contribute to shortening the time required to perform the inspection operation of the first transfer contact **221** and the second transfer contact **222**. Meanwhile, the portion of the second transmission contact **222** on which the second transmission inspection plane **2222** is formed and the portion of the first transmission contact **221** on which the first transmission inspection plane **2212** is formed may have the same function.

(82) As illustrated in FIGS. **10** and **11**, the second transmission inspection plane **2222**, the first transmission inspection plane **2212**, the second RF inspection plane **2122** and the first RF inspection plane **2112** may be disposed on a plane having the same height. Accordingly, since the height at which the inspection instrument is raised or lowered to contact each of the second transmission inspection plane **2222**, the first transmission inspection plane **2212**, the second RF inspection plane **2122** and the first RF inspection plane **2112** may be implemented to be the same, the substrate connector **1** according to the first example may further improve the easy and accuracy of the inspection operation of the second transmission contact **222**, the first transmission contact **221**, the second RF contact **212** and the first RF contact **211**. Further, in the substrate connector **1** according to the first example, since the inspection instrument can simultaneously contact the second transmission inspection plane **2222**, the first transmission inspection plane **2212**, the second RF inspection plane **2122** and the first RF inspection plane **2112**, it may contribute to further shortening the time taken to perform the inspection operation of the second transmission contact **222**, the first transmission contact **221**, the second RF contact **212** and the first RF contact **211**.

(83) As illustrated in FIGS. **12** and **13**, the first RF inspection plane **2112** may be disposed at a higher position than the first transmission inspection planes **2212**. In this case, the first RF inspection plane **2112** and the second RF inspection plane **2122** may be disposed on a plane having the same height. The first transmission inspection planes **2212** and the second transmission inspection planes **2222** may be disposed on a plane having the same height, and may be disposed in a lower position than the first RF inspection plane **2112** and the second RF inspection plane **2122**. Accordingly, the substrate connector **1** according to the first example may be suitably implemented such that an inspection operation in which the inspection instrument simultaneously inspects the first RF contact **211** and the second RF contact **212** and an inspection operation in which the inspection instrument simultaneously inspects the first transmission contacts **221** and the second transmission contacts **222** are individually performed. Although not illustrated, the first RF inspection plane **2112** may be disposed at a lower position than the first transmission inspection planes **2212**. In this case, the first transmission inspection planes **2212** and the second transmission inspection planes **2222** may be disposed on a plane having the same height, and may be disposed at a higher position than the first RF inspection plane **2112** and the second RF inspection plane **2122**.

(84) As illustrated in FIGS. **5** and **6**, the first RF inspection plane **2112** and the first transmission inspection plane **2212** may be disposed on a first row **R1** which is parallel to the first axial direction (X-axis direction). The second RF inspection plane **2122** and the second transmission inspection planes **2212** may be disposed on a second row **R2** which is parallel to the first axial direction (X-axis direction). The first row **R1** and the second row **R2** may be disposed to be spaced apart from each other in the second axial direction (Y-axis direction). Accordingly, the substrate connector **1** according to the first example may be suitably implemented such that an inspection

operation in which the inspection instrument simultaneously inspects the first RF contact **211** and the first transmission contacts **221** and an inspection operation in which the inspection instrument simultaneously inspects the second RF contact **212** and the second transmission contacts **222** are individually performed. In addition, the substrate connector **1** according to the first example may be suitably implemented such that an inspection operation in which the inspection instrument simultaneously inspects the first RF contact **211**, the first transmission contacts **221**, the second RF contact **212** and the second transmission contacts **222** is performed. Accordingly, the substrate connector **1** according to the first example is implemented such that the inspection instrument can inspect the first RF contact **211**, the first transmission contacts **221**, the second RF contact **212** and the second transmission contacts **222** in various inspection methods according to the inspection conditions, inspection environment and the like.

(85) As illustrated in FIGS. **5**, **9** and **14**, the first transmission inspection planes **2212** may be disposed on the first row **R1**. The second transmission inspection planes **2222** may be disposed on the second row **R2**. The first RF inspection plane **2112** and the second RF inspection plane **2122** may be disposed on a third row **R3** which is parallel to the first axial direction (X-axis direction). The first row **R1**, the second row **R2** and the third row **R3** may be disposed to be spaced apart from each other in the second axial direction (Y-axis direction). Accordingly, the substrate connector **1** according to the first example is implemented such that the inspection instruments can inspect the first RF contact, the first transmission contacts **221**, the second RF contact **212** and the second transmission contacts **222** in more various inspection methods according to the inspection conditions, inspection environment and the like. FIG. **14** illustrates that the third row **R3** is disposed in the middle on the basis of the second axial direction (Y-axis direction), but is not limited thereto, and the first row **R1** or the second row **R2** may be disposed in the middle.

(86) Next, the first ground contact **250** may be implemented as follows.

(87) Referring to FIGS. **2** to **14**, the first ground contact **250** may include a 1-1 ground contact **251** and a 1-2 ground contact **252**.

(88) The 1-1 ground contact **251** is disposed between the first RF contact **211** and the first transmission contacts **221** on the basis of the first axial direction (X-axis direction). Accordingly, the 1-1 ground contact **251** may shield between the first RF contact **211** and the first transmission contacts **221**.

(89) The 1-1 ground contact **251** may include a 1-1 ground inspection plane **2511**.

(90) The 1-1 ground inspection plane **2511** is for the inspection instrument to come into contact. The inspection instrument may perform an inspection on the 1-1 ground contact **251** in a state of being in contact with the 1-1 ground inspection plane **2511**. The 1-1 ground inspection plane **2511** may be formed as a flat surface. Accordingly, since the substrate connector **200** according to the first example may secure an area for the inspection instrument to contact the 1-1 ground contact **251** using the 1-1 ground inspection plane **2511**, it is possible to improve the ease and accuracy of the contact between the inspection instrument and the 1-1 ground contact **251**. Accordingly, in the substrate connector **200** according to the first example, the accuracy of the inspection result for the 1-1 ground contact **251** may be improved by using the 1-1 ground inspection plane **2511**. Further, in comparison with the comparative example in which the surface of the 1-1 ground contact **251** for contacting the inspection instrument is formed as a curved surface, since the 1-1 ground inspection plane **2511** is formed as a flat surface, the substrate connector **200** according to the first example is implemented to prevent slip or the like from occurring while the inspection instrument is in contact with the 1-1 ground inspection plane **2511**. Accordingly, the substrate connector **1** according to the first example may improve the contact reliability between the inspection instrument and the 1-1 ground contact **251**.

(91) When the width **H** (illustrated in FIGS. **7** and **8**) on the basis of the first axial direction (X-axis direction) of the 1-1 ground inspection plane **2511** is 1, it may be formed as a plane having the length **L** (illustrated in FIGS. **7** and **8**) of 0.5 or more on the basis of the second axial direction (Y-

axis direction). When the width H (illustrated in FIGS. 7 and 8) on the basis of the first axial direction (X-axis direction) of the 1-1 ground inspection plane **2511** is 1, it may be formed as a plane having the length L (illustrated in FIGS. 7 and 8) of 1.5 or less on the basis of the second axial direction (Y-axis direction). When the width H (illustrated in FIGS. 7 and 8) on the basis of the first axial direction (X-axis direction) of the 1-1 ground inspection plane **2511** is 1, it may be formed as a plane having the length L (illustrated in FIGS. 7 and 8) of 0.5 or more to 1.5 or less on the basis of the second axial direction (Y-axis direction). That is, the 1-1 ground inspection plane **2511** may be formed as a plane in which the ratio of the length L to the width H is 0.5 or more to 1.5 or less. Accordingly, the substrate connector **200** according to the first example may not only secure an area for the inspection instrument to contact the 1-1 ground contact **251** using the 1-1 ground inspection plane **2511**, but also it is possible to realize miniaturization by reducing the overall size.

(92) The 1-1 ground contact **251** may include a 1-1 ground mounting member **2512**.

(93) The 1-1 ground mounting member **2512** is for mounting on the first substrate. The 1-1 ground contact **251** may be electrically connected to the first substrate through the 1-1 ground mounting member **2512**. The 1-1 ground mounting member **2512** may be formed to have a length that can be connected to the ground housing **230** based on the second axis direction (Y-axis direction). In this case, the 1-1 ground mounting member **2512** may be connected to a sidewall of the ground housing **230**. The 1-1 ground mounting member **2512** may be formed in a plate shape disposed in the horizontal direction.

(94) The 1-1 ground contact **251** may include a 1-1 ground connecting member **2513**.

(95) The 1-1 ground connecting member **2513** is for connecting to a ground contact of the counterpart connector. As the 1-1 ground connecting member **2513** is connected to a ground contact of the counterpart connector, the 1-1 ground contact **251** may be electrically connected to a transmission contact of the counterpart connector. Accordingly, the shielding power of the 1-1 ground contact **251** with respect to the first RF contacts **211** may be strengthened. The 1-1 ground connecting member **2513** may be connected to the 1-1 ground mounting member **2512**. The 1-1 ground connecting member **2513** may be directly coupled to the 1-1 ground mounting member **2512**. The 1-1 ground connecting member **2513** may be coupled to another member of the 1-1 ground connecting member **2513** and may be connected to the 1-1 ground mounting member **2512** through the member. The 1-1 ground connecting member **2513** may be formed in a plate shape disposed in a vertical direction. The 1-1 ground connecting member **2513** may be implemented in a form where a plate material disposed in a vertical direction and a plate material disposed in a horizontal direction are combined through bending processing of the plate material. Meanwhile, the 1-1 ground inspection plane **2511** may be formed on the 1-1 ground mounting member **2512** or the 1-1 ground connecting member **2513**.

(96) The 1-1 ground contact **251** may include a 1-1 ground inspection member **2514**.

(97) The 1-1 ground inspection member **2514** is for contacting the inspection instrument. In this case, as illustrated in FIG. 6, the 1-1 ground inspection plane **2511** may be formed on the 1-1 ground inspection member **2514**. The 1-1 ground inspection plane **2511** may be formed on a surface of the 1-1 ground inspection member **2514** facing the inner space **230a**. The 1-1 ground inspection member **2514** may be coupled to each of the 1-1 ground connecting member **2513** and the 1-1 ground mounting member **2512**. In this case, the 1-1 ground inspection member **2514** has a connection function for connecting the 1-1 ground connecting member **2513** and the 1-1 ground mounting member **2512** and a contact function for the inspection instrument. The 1-1 ground inspection member **2514** may be coupled to the 1-1 ground connecting member **2513** or the 1-1 ground mounting member **2512**. In this case, the 1-1 ground inspection member **2514** may have a support function supported by the insulation unit **240** and a contact function for the inspection instrument. Although not illustrated, even when the 1-1 ground inspection member **2514** is provided, the 1-1 ground inspection plane **2511** may be formed on the 1-1 ground mounting

member **2512** or the 1-1 ground connecting member **2513**. In this case, the 1-1 ground inspection member **2514** may have a support function for the insulation unit **240** or a connection function for connecting the 1-1 ground connecting member **2513** and the 1-1 ground mounting member **2512**. (98) Meanwhile, when the 1-1 ground inspection plane **2511** is formed in a portion of the 1-1 ground contact **251** that is covered by the insulation unit **240**, an inspection connection window (not illustrated) may be formed on the insulation unit **240**. Accordingly, the 1-1 ground inspection plane **2511** may be disposed to be exposed to the inner space **230a** without being covered by the insulation unit **240** through the inspection connection window.

(99) Next, the 1-2 ground contact **252** may be implemented as follows.

(100) Referring to FIGS. **2** to **14**, the 1-2 ground contact **252** may include a 1-2 ground inspection plane **2521**. The 1-2 ground inspection plane **2521** is for the inspection instrument to contact. Since the 1-2 ground inspection plane **2521** is implemented to substantially coincide with the 1-1 ground inspection plane **2511** of the 1-1 ground contact **251** as described above, the detailed description thereof will be omitted.

(101) The 1-2 ground contact **252** may include a 1-2 ground mounting member **2522**. The 1-2 ground mounting member **2522** is for mounting on the first substrate. Since the 1-2 ground mounting member **2522** is implemented to substantially coincide with the 1-1 ground mounting member **2512** of the 1-1 ground contact **251** as described above, the detailed description thereof will be omitted.

(102) The 1-2 ground contact **252** may include a 1-2 ground connecting member **2523**. The 1-2 ground connecting member **2523** is for being connected to a ground contact of the counterpart connector. Since the 1-2 ground connecting member **2523** is implemented to substantially coincide with the 1-1 ground connecting member **2513** of the 1-1 ground contact **251** as described above, the detailed description thereof will be omitted. The 1-2 ground inspection plane **2521** may be formed on the 1-2 ground mounting member **2522** or the 1-2 ground connecting member **2523**.

(103) The 1-2 ground contact **252** may include a 1-2 ground inspection member **2524**. The 1-2 ground inspection member **2524** is for contacting the inspection instrument. Since the 1-2 ground inspection member **2524** is implemented to substantially coincide with the 1-1 ground inspection member **2514** of the 1-1 ground contact **251** as described above, the detailed description thereof will be omitted. The 1-2 ground inspection plane **2521** may be formed on any one of the 1-2 ground inspection member **2524**, the 1-2 ground connecting member **2523** and the 1-2 ground mounting member **2522**.

(104) As illustrated in FIGS. **10** to **13**, the 1-2 ground inspection plane **2521** and the 1-1 ground inspection plane **2511** may be disposed on a plane having the same height. Accordingly, since the height at which the inspection instrument is raised and lowered to contact each of the 1-2 ground inspection plane **2521** and the 1-1 ground inspection plane **2511** may be implemented to be the same, the substrate connector **1** according to the first example may further improve the ease and accuracy of the inspection operation of the 1-1 ground contact **251** and the 1-2 ground contact **252**. Further, in the substrate connector **1** according to the first example, since it is possible for the inspection instrument to contact the 1-2 ground inspection plane **2521** and the 1-1 ground inspection plane **2511** at the same time, it may contribute to shortening the time taken to perform the inspection operation of the 1-1 grounded contact **251** and the 1-2 grounded contact **252**. Meanwhile, the portion of the 1-2 ground contact **252** on which the 1-2 ground inspection plane **2521** is formed and the portion of the 1-1 ground contact **251** on which the 1-1 ground inspection plane **2511** is formed may have the same function as each other.

(105) As illustrated in FIGS. **10** and **11**, the 1-2 ground inspection plane **2521**, the 1-1 ground inspection plane **2511**, the second RF inspection plane **2122**, the first RF inspection plane **2112**, the second transmission inspection planes **2222** and the first transmission inspection planes **2212** may be disposed on a plane having the same height. Accordingly, since the height at which the inspection instrument is raised or lowered to be in contact with each of the 1-2 ground inspection

plane **2521**, the 1-1 ground inspection plane **2511**, the second RF inspection plane **2122**, the first RF inspection plane **2112**, the second transmission inspection planes **2222** and the first transmission inspection planes **2212** may be implemented to be the same, the substrate connector **1** according to the first example may further improve the ease and accuracy of the inspection operation for the 1-2 ground contact **252**, the 1-1 ground contact **251**, the second RF contact **212**, the first RF contact **211**, the second transmission contacts **222** and the first transmission contacts **221**. Further, in the substrate connector **1** according to the first example, since it is possible for the inspection instrument to simultaneously contact the 1-1 ground inspection plane **2521**, the 1-1 ground inspection plane **2511**, the second RF inspection plane **2122**, the first RF inspection plane **2112**, the second transmission inspection plane **2222** and the first transmission inspection planes **2212**, it may contribute to shortening the time taken to perform the inspection operation for the 1-2 ground contact **252**, the 1-1 ground contact **251**, the second RF contact **212**, the first RF contact **211**, the second transmission contacts **222** and the first transmission contacts **221**.

(106) As illustrated in FIGS. **12** and **13**, the first RF inspection plane **2112** and the second RF inspection plane **2122** may be disposed at a higher position than the 1-1 ground inspection plane **2511** and the 1-2 ground inspection plane **2521**. Although not illustrated, the first RF inspection plane **2112** and the second RF inspection plane **2122** may be disposed at a lower position than the 1-1 ground inspection plane **2511** and the 1-2 ground inspection plane **2521**.

(107) As illustrated in FIGS. **5** and **6**, the 1-1 ground inspection plane **2511** and the first transmission inspection planes **2212** may be disposed on the first row **R1**. The 1-2 ground inspection plane **2521** and the second transmission inspection plane **2222** may be disposed on the second row **R2**. Accordingly, the substrate connector **1** according to the first example is implemented such that the inspection instrument can inspect the 1-1 ground contact **251**, the first transmission contacts **221**, the 1-2 ground contact **252** and the second transmission contacts **222** in various inspection methods according to the inspection conditions, inspection environment and the like.

(108) In this case, the first RF inspection plane **2112** may be disposed on the first row **R1**, and the second RF inspection plane **2122** may be disposed on the second row **R2**. Accordingly, the substrate connector **1** according to the first example is implemented such that the inspection instrument can inspect the first RF contact **211**, the 1-1 ground contact **251**, the first transmission contacts **221**, the second RF contact **212**, the 1-2 ground contact **252** and the second transmission contacts **222** in various inspection methods according to the inspection conditions, inspection environment and the like.

(109) As illustrated in FIGS. **5**, **9** and **14**, the first RF inspection plane **2112** and the second RF inspection plane **2122** may be disposed on the third row **R3**. Accordingly, the substrate connector **1** according to the first example is implemented such that the inspection instrument can inspect the first RF contact **211**, the 1-1 ground contact **251**, the first transmission contacts **221**, the second RF contact **212**, the 1-1 ground contact **252** and the second transmission contacts **222** in more various inspection methods according to the inspection conditions, inspection environment and the like.

(110) Next, the second ground contact **260** may be implemented as follows.

(111) Referring to FIGS. **2** to **14**, the second ground contact **260** may include a 2-1 ground contact **261** and a 2-2 ground contact **262**.

(112) The 2-1 ground contact **261** is disposed between the second RF contact **212** and the first transmission contacts **221** on the basis of the first axial direction (X-axis direction). Accordingly, the 2-1 ground contact **261** may shield between the second RF contact **212** and the first transmission contact **221**.

(113) The 2-1 ground contact **261** may include a 2-1 ground inspection plane **2611**, a 2-1 ground mounting member **2612** and a 2-1 ground connecting member **2613**. The 2-1 ground contact **261** may further include a 2-1 ground inspection member **2614**. The 2-1 ground inspection plane **2611** may be formed on any one of the 2-1 ground mounting member **2612**, the 2-1 ground connecting

member **2613** and the 2-1 ground inspection member **2614**. Since the 2-1 ground inspection plane **2611**, the 2-1 ground mounting member **2612**, the 2-1 ground connecting member **2613** and the 2-1 ground inspection member **2614** may be implemented to respectively coincide with the 1-1 ground inspection plane **2511**, the 1-1 ground mounting member **2512**, the 1-1 ground connecting member **2513** and the 1-1 ground inspection member **2514**, the detailed descriptions thereof will be omitted. (114) The 2-1 ground contact **261** and the 1-1 ground contact **251** may be formed in the same shape as each other. Accordingly, the substrate connector **200** according to the first example may improve the ease of manufacturing operations for manufacturing each of the 2-1 ground contact **261** and the 1-1 ground contact **251**. In this case, the 2-1 ground contact **261** and the 1-1 ground contact **251** may be arranged to be point-symmetric with respect to a symmetry point SP (illustrated in FIGS. **6** and **9**). The symmetry point SP is a point which is spaced apart by the same distance from each of both side walls **230b**, **230c** (illustrated in FIG. **15**) of the ground housing **230** disposed to be spaced apart from each other on the basis of the first axial direction (X-axis direction), and additionally is spaced apart by the same distance from each of both side walls **230d**, **230e** (illustrated in FIG. **15**) of the ground housing **230** disposed to be spaced apart from each other on the basis of the second axial direction (Y-axis direction). Therefore, since the substrate connector **200** according to the first example is implemented such that the 2-1 ground contact **261** and the 1-1 ground contact **251** are formed in the same shape as each other with only different arrangement directions, the ease of manufacturing operations for manufacturing the 2-1 ground contact **261** and the 1-1 ground contact **251** may be further improved. In this case, the second RF contact **212** and the first RF contact **211** may be arranged to be point-symmetric with respect to the symmetry point SP.

(115) The second 2-2 ground contact **262** is disposed between the second RF contact **212** and the second transmission contacts **222** on the basis of the first axial direction (X-axis direction). Accordingly, the 2-2 ground contact **262** may shield between the second RF contact **212** and the second transmission contacts **222**.

(116) The 2-2 ground contact **262** may include a 2-2 ground inspection plane **2621**, a 2-2 ground mounting member **2622** and a 2-2 ground connecting member **2623**. The 2-2 ground contact **262** may further include a 2-2 ground inspection member **2624**. The 2-2 ground inspection plane **2621** may be formed on any one of the 2-2 ground mounting member **2622**, the 2-2 ground connecting member **2623** and the 2-2 ground inspection member **2624**. Since the 2-2 ground inspection plane **2621**, the 2-2 ground mounting member **2622**, the 2-2 ground connecting member **2623** and the 2-2 ground inspection member **2624** may be implemented to respectively substantially coincide with the 1-2 ground inspection plane **2521**, the 1-2 ground mounting member **2522**, the 1-2 ground connecting member **2523** and the 1-2 ground inspection member **2524**, the detailed descriptions thereof will be omitted.

(117) The 2-2 ground contact **262** and the 1-2 ground contact **252** may be formed to in the same shape as each other. Accordingly, the substrate connector **200** according to the first example may improve the ease of manufacturing operations of manufacturing each of the 2-2 ground contact **262** and the 1-2 ground contact **252**. In this case, the 2-2 ground contact **262** and the 1-2 ground contact **252** may be arranged to be point-symmetrical with respect to the symmetry point SP (illustrated in FIGS. **6** and **9**) as a reference. The 2-2 ground contact **262**, the 1-2 ground contact **252**, the 2-1 ground contact **261** and the 1-1 ground contact **251** may all be formed in the same shape as each other.

(118) Referring to FIGS. **2** to **16**, in the substrate connector **200** according to the first example, the ground housing **230** may be implemented as follows.

(119) The ground housing **230** may include a ground inner wall **231**, an outer ground wall **232** and a ground connection wall **233**.

(120) The ground inner wall **231** faces the insulation unit **240**. The ground inner wall **231** may be disposed to face the inner space **230a**. The 1-1 ground contact **251** and the 2-1 ground contact **261** may be respectively connected to the inner ground wall **231**. The ground inner wall **231** may be

disposed to surround all sides with respect to the inner space **230a**. Although not illustrated, the ground inner wall **231** may include a plurality of sub-ground inner walls such that the sub-ground inner walls are implemented to be disposed on different sides with respect to the inner space **230a**. In this case, the sub-ground inner walls may be spaced apart from each other.

(121) The ground inner wall **231** may be connected to a ground housing of a counterpart connector inserted into the inner space **230a**. For example, as illustrated in FIG. **16**, the ground inner wall **231** may be connected to the ground housing **330** of the counterpart connector. In this way, the substrate connector **200** according to the first example may further strengthen the shielding function through the connection between the ground housing **230** and the ground housing of the counterpart connector. In addition, the substrate connector **200** according to the first example may reduce electrical adverse effects such as crosstalk and the like that may be generated by capacitance or induction between adjacent terminals through a connection between the ground housing **230** and the ground housing of the counterpart connector. In this case, since the substrate connector **200** according to the first example may secure a path through which electromagnetic waves are introduced into at least one ground of the first substrate and the second substrate, the EMI shielding performance may be further strengthened.

(122) The ground outer wall **232** is spaced apart from the ground inner wall **231**. The ground outer wall **232** may be disposed outside the ground inner wall **231**. The ground outer wall **232** may be disposed to surround all sides with respect to the ground inner wall **231**. The ground outer wall **232** and the ground inner wall **231** may be implemented as a shielding wall surrounding the side of the inner space **230a**. The first RF contact **211** and the second RF contact **212** may be located in the inner space **230a** surrounded by the shielding wall. Accordingly, the ground housing **230** may implement a shielding function for the RF contacts **210** using a shielding wall. Therefore, the substrate connector **200** according to the first example may contribute to further improving EMI shielding performance and EMC performance by using the shielding wall.

(123) The ground outer wall **232** may be grounded by being mounted on the first substrate. In this case, the ground housing **230** may be grounded through the ground outer wall **232**. When one end of the ground outer wall **232** is coupled to the ground connection wall **233**, the other end of the ground outer wall **232** may be mounted on the first substrate. In this case, the ground outer wall **232** may be formed to have a higher height than the ground inner wall **231**.

(124) The ground connection wall **233** is coupled to each of the ground inner wall **231** and the ground outer wall **232**. The ground connection wall **233** may be disposed between the ground inner wall **231** and the ground outer wall **232**. The ground inner wall **231** and the ground outer wall **232** may be electrically connected to each other through the ground connection wall **233**. Accordingly, when the grounding outer wall **232** is mounted on the first substrate and grounded, the grounding connection wall **233** and the ground inner wall **231** are also grounded to implement a shielding function.

(125) The ground connection wall **233** may be coupled to one end of the ground outer wall **232** and one end of the ground inner wall **231**, respectively. Based on FIG. **16**, one end of the grounded outer wall **232** may correspond to the upper end of the grounded outer wall **232**, and one end of the grounded inner wall **231** may correspond to the upper end of the grounded inner wall **231**. The ground connection wall **233** may be formed in a plate shape disposed in a horizontal direction, and the ground outer wall **232** and the ground inner wall **231** may be formed in a plate shape disposed in a vertical direction, respectively. The ground connection wall **233**, the ground outer wall **232** and the ground inner wall **231** may be integrally formed.

(126) The ground connection wall **233** may be connected to a ground housing of a counterpart connector inserted into the inner space **230a**. Accordingly, in the substrate connector **200** according to the first example, since the ground outer wall **232** and the ground connection wall **233** are connected to the ground housing of the counterpart connector, the shielding function may be further strengthened by increasing the contact area between the ground housing **230** and the ground

housing of the counterpart connector.

(127) The ground floor **234** protrudes from the lower end of the ground inner wall **231** toward the inner space **230a**. That is, the ground floor **234** may protrude to the inside of the ground inner wall **231**. The ground floor **234** may extend along the lower end of the ground inner wall **231** to be formed in a closed ring shape. The ground floor **234** may be grounded by being mounted on the first substrate. In this case, the ground housing **330** may be grounded through the ground floor **234**. When the counterpart connector is inserted into the inner space **230a**, the grounding floor **234** may be connected to a grounding housing of the counterpart connector. The ground floor **234** may be formed in a plate shape arranged in a horizontal direction.

(128) Herein, the ground housing **230** may implement a shielding function for the first RF contact **211** together with the first ground contact **250**. The ground housing **230** may implement a shielding function for the second RF contact **212** together with the second ground contact **260**.

(129) In this case, as illustrated in FIG. **15**, the ground housing **230** may include a first shielding wall **230b**, a second shielding wall **230c**, a third shielding wall **230d** and a fourth shielding wall **230e**. The first shielding wall **230b**, the second shielding wall **230c**, the third shielding wall **230d** and the fourth shielding wall **230e** may be respectively implemented by the ground inner wall **231**, the ground outer wall **232** and the ground connection wall **233**. The first shielding wall **230b** and the second shielding wall **230c** are disposed to face each other on the basis of the first axial direction (X-axis direction). The first RF contact **211** and the second RF contact **212** may be positioned between the first shielding wall **230b** and the second shielding wall **230c** on the basis of the first axial direction (X-axis direction). Based on the first axial direction (X-axis direction), the first RF contact **211** may be located at a position where the distance away from the first shielding wall **230b** is shorter than the distance away from the second shielding wall **230c**. Based on the first axial direction (X-axis direction), the second RF contact **212** may be located at a position where the distance away from the second shielding wall **230c** is shorter than the distance away from the first shielding wall **230b**. The third shielding wall **230d** and the fourth shielding wall **230e** are disposed to face each other on the basis of the second axial direction (Y-axis direction). The first RF contact **211** and the second RF contact **212** may be positioned between the third shielding wall **230d** and the fourth shielding wall **230e** on the basis of the second axial direction (Y-axis direction).

(130) The first ground contact **250** may be positioned between the first RF contact **211** and the transmission contacts **220** on the basis of the first axial direction (X-axis direction). Accordingly, the first RF contact **211** may be positioned between the first shielding wall **230b** and the first ground contact **250** on the basis of the first axial direction (X-axis direction), and may be positioned between the third shielding wall **230d** and the fourth shielding wall **230e** on the basis of the second axial direction (Y-axis direction). Accordingly, the substrate connector **200** according to the first example may use the first ground contact **250**, the first shielding wall **230b**, the third shielding wall **230d** and the fourth shielding wall **230e** to strengthen the shielding function for the first RF contact **211**. The first ground contact **250**, the first shielding wall **230b**, the third shielding wall **230d** and the fourth shielding wall **230e** may be disposed on four sides with respect to the first RF contact **211** to implement the shielding power against RF signals. In this case, the first ground contact **250**, the first shielding wall **230b**, the third shielding wall **230d** and the fourth shielding wall **230e** may implement the first ground loop **250a** (illustrated in FIG. **15**) for the first RF contact **211**. Therefore, the substrate connector **200** according to the first example may further strengthens the shielding function for the first RF contact **211** by using the first ground loop **250a** to realize complete shielding for the first RF contact **211**.

(131) The second ground contact **260** may be positioned between the second RF contact **212** and the transmission contacts **220** on the basis of the first axial direction (X-axis direction). Accordingly, the second RF contact **212** may be positioned between the second shielding wall **230c** and the second ground contact **260** on the basis of the first axial direction (X-axis direction), and may be positioned between the third shielding wall **230d** and the fourth shielding wall **230e** on the

basis of the second axial direction (Y-axis direction). Accordingly, the substrate connector **200** according to the first example may use the second ground contact **260**, the second shielding wall **230c**, the third shielding wall **230d** and the fourth shielding wall **230e** to strengthen the shielding function for the second RF contact **212**. The second ground contact **260**, the second shielding wall **230c**, the third shielding wall **230d** and the fourth shielding wall **230e** may be disposed on four sides with respect to the second RF contact **212** to implement the shielding power against RF signals. In this case, the second ground contact **260**, the second shielding wall **230c**, the third shielding wall **230d** and the fourth shielding wall **230e** may implement the second ground loop **260a** (illustrated in FIG. 15) for the second RF contact **212**. Therefore, the substrate connector **200** according to the first example may further strengthen the shielding function for the second RF contact **212** using the second ground loop **260a**, to realize complete shielding for the second RF contact.

(132) Referring to FIGS. 2 to 16, in the substrate connector **200** according to the first example, the insulation unit **240** may be implemented as follows.

(133) The insulation unit **240** may include an insulating member **241**, an insertion member **242** and a connecting member **243**.

(134) The insulating member **241** supports the RF contacts **210** and the transmission contacts **220**. The insulating member **241** may be positioned in the inner space **230a**. The insulating member **241** may be positioned inside the ground inner wall **231**. The insulating member **241** may be inserted into an inner space of the counterpart connector.

(135) The insertion member **242** is inserted between the ground inner wall **231** and the ground outer wall **232**. As the insertion member **242** is inserted between the ground inner wall **231** and the ground outer wall **232**, the insulation unit **240** may be coupled to the ground housing **230**. The insertion member **242** may be inserted between the ground inner wall **231** and the grounding outer wall **232** by the interference fit method. The insertion member **242** may be disposed outside the insulating member **241**. The insertion member **242** may be disposed to surround the outside of the insulating member **241**.

(136) The connecting member **243** is coupled to each of the insertion member **242** and the insulating member **241**. The insertion member **242** and the insulating member **241** may be connected to each other through the connecting member **243**. Based on the vertical direction, the connecting member **243** may be formed to have a thinner thickness than that of the inserting member **242** and the insulating member **241**. Accordingly, a space is provided between the insertion member **242** and the insulating member **241**, and the counterpart connector may be inserted into the space. The connecting member **243**, the inserting member **242** and the connecting member **243** may be integrally formed.

(137) The insulation unit **240** may include a soldering inspection window **244** (illustrated in FIG. 4).

(138) The soldering inspection window **244** may be formed through the insulation unit **240**. The soldering inspection window **244** may be used to inspect a state in which the RF mounting members **2111**, **2121** are mounted on the first substrate. In this case, the RF contacts **210** may be coupled to the insulation unit **240** such that the RF mounting members **2111**, **2121** are positioned on the soldering inspection window **244**. Accordingly, the RF mounting members **2111**, **2121** are not covered by the insulation unit **240**. Therefore, in a state where the substrate connector **200** according to the first example is mounted on the first board, the operator may inspect the mounted state of the RF mounting members **2111**, **2121** on the first substrate through the soldering inspection window **244**. Accordingly, in the substrate connector **200** according to the first example, even if all of the RF contacts **210** including the RF mounting members **2111**, **2121** are positioned inside the ground housing **230**, it is possible to improve the accuracy of the mounting operation of mounting the RF contacts **210** on the first substrate. The soldering inspection window **244** may be formed through the insulating member **241**.

(139) The insulation unit **240** may include a plurality of the soldering inspection windows **244**. In this case, the RF mounting members **2111**, **2121** may be positioned in different soldering inspection windows **244**. The transmission mounting members **2211**, **2221** may be positioned in some of the soldering inspection windows **244**. The ground mounting members **2512**, **2522**, **2612**, **2622** may be positioned in some of the soldering inspection windows **244**. Therefore, in a state where the substrate connector **200** according to the first example is mounted on the first substrate, the operator may inspect the mounting state of the RF mounting members **2111**, **2121**, the transmission mounting members **2211**, **2221** and the ground mounting members **2512**, **2522**, **2612**, **2622** on the first substrate through the soldering inspection windows **244**. Accordingly, the substrate connector **200** according to the first example may improve the accuracy of the operation of mounting the RF mounting members **2111**, **2121**, the transmission mounting members **2211**, **2221** and the ground mounting members **2512**, **2522**, **2612**, **2622** on the first substrate. The soldering inspection windows **244** may be formed to pass through the insulation unit **240** at positions spaced apart from each other.

Substrate Connector 300 According to the Second Example

(140) Referring to FIGS. **2** to **20**, the substrate connector **300** according to the second example may be mounted on the second substrate. When the substrate connector **300** according to the second example and the counterpart connector are assembled to be coupled to each other, the second substrate on which the substrate connector **300** according to the second example is mounted and the first substrate on which the counterpart connector is mounted may be electrically connected to each other. In this case, the counterpart connector may be implemented as the substrate connector **200** according to the first example. Meanwhile, the counterpart connector in the substrate connector **200** according to the first example may be implemented as the substrate connector **300** according to the second example.

(141) The substrate connector **300** according to the second example may include a plurality of RF contacts **310**, a plurality of transmission contacts **220**, a ground housing **330**, an insulation unit **340**, a first ground contact **350** and a second ground contact **360**. Since the RF contacts **310**, the transmission contacts **220**, the ground housing **330**, the insulation unit **340**, the first ground contact **350** and the second ground contact **360** may be implemented to substantially coincide with the RF contacts **210**, the transmission contacts **220**, the ground housing **230**, the insulation unit **240**, the first ground contact **250** and the second ground contact **260** in the substrate connector **200** according to the first example as described above, respectively, the differences will be mainly described below.

(142) A first RF contact **311** of the RF contacts **310** and a second RF contact **312** of the RF contacts **310** may be supported in the insulation unit **340** at a position which is spaced apart from each other in the first axial direction (X-axis direction).

(143) The first RF contact **311** includes a first RF mounting member **3111** for mounting on the second substrate, a first RF inspection plane **3112** for contacting the inspection instrument and a first RF connecting member **3113** for contacting an RF contact of the counterpart connector. The first RF contact **311** may include a plurality of first RF mounting members **3111**, **3111'** (illustrated in FIG. **20**). The first RF contact **311** may further include a first RF inspection member **3114** for contacting the inspection instrument. Since the first RF mounting member **3111**, the first RF inspection plane **3112**, the first RF connecting member **3113** and the first RF inspection member **3114** may be implemented to substantially coincide with the first RF mounting member **2111**, the first RF inspection plane **2112**, the first RF connecting member **2113** and the first RF inspection member **2114** in the substrate connector according to the first example as described above, respectively, the detailed descriptions thereof will be omitted.

(144) The second RF contact **312** may include a second RF mounting member **3121** for mounting on the second substrate, a second RF inspection plane **3122** for contacting the inspection instrument, and a second RF connecting member **3123** for connecting to an RF contact of the

counterpart connector. The second RF contact **312** may include a plurality of second RF mounting members **3121**, **3121'** (illustrated in FIG. 20). The second RF contact **312** may further include a second RF inspection member **3124** for contacting the inspection instrument. Since the second RF mounting member **3121**, the second RF inspection plane **3122**, the second RF connecting member **3123** and the second RF inspection member **3124** may be implemented to substantially coincide with the second RF mounting member **2121**, the second RF inspection plane **2122**, the second RF connecting member **2123** and the second RF inspection member **2124** in the substrate connector **200** according to the first example as described above, respectively, the detailed descriptions thereof will be omitted.

(145) The transmission contacts **220** may be disposed between the first RF contact **311** and the second RF contact **312** on the basis of the first axial direction (X-axis direction). The first transmission contacts **321** of the transmission contacts **220** and the second transmission contacts **322** of the transmission contacts **220** may be disposed to be spaced apart from each other in the second axial direction (Y-axis direction).

(146) The first transmission contacts **321** may be disposed to be spaced apart from each other in the first axial direction (X-axis direction). Each of the first transmission contacts **321** may include a first transmission mounting member **3211** for mounting on the second substrate, a first transmission inspection plane **3212** for contacting the inspection instrument, and a first transmission connecting member **3212** for connecting to a transmission contact of the counterpart connector. Each of the first transmission contacts **321** may further include a first transmission inspection member **3214** for contacting the inspection instrument. Since the first transmission mounting member **3211**, the first transmission inspection plane **3212**, the first transmission connecting member **3213** and the first transmission inspection member **3214** may be implemented to substantially coincide with the first transmission mounting member **2211**, the first transmission inspection plane **2212**, the first transmission connecting member **2213** and the first transmission inspection member **2214** in the substrate connector **200** according to the first example as described above, respectively, the detailed descriptions thereof will be omitted.

(147) The second transmission contacts **322** may be disposed to be spaced apart from each other in the first axial direction (X-axis direction). Each of the second transmission contacts **322** may include a second transmission mounting member **3221** for mounting on the second substrate, a second transmission inspection plane **3222** for contacting the inspection instrument, and a second transmission connecting member **3223** for connecting to a transmission contact of the counterpart connector. The second transmission contacts **322** may further include a second transmission inspection member **3224** for contacting the inspection instrument. Since the second transmission mounting member **3221**, the second transmission inspection plane **3222**, the second transmission connecting member **3223** and the second transmission inspection member **3224** may be implemented to substantially coincide with the second transmission mounting member **2221**, the second transmission inspection plane **2222**, the second transmission connecting member **2223** and the second transmission inspection member **2224** in the substrate connector **200** according to the first example as described above, respectively, the detailed descriptions thereof will be omitted.

(148) The ground housing **330** is coupled to the insulation unit **340**. The ground housing **330** may be grounded by being mounted on the second substrate. The ground housing **330** may be disposed to surround the sides of the inner space **330a**. The insulation unit **340** may be positioned in the inner space **330a**. The first RF contact **311**, the second RF contact **312**, the transmission contacts **320**, the first ground contact **250** and the second ground contact **260** may all be positioned in the inner space **330a**. In this case, in each of the first RF contact **311**, the second RF contact **312**, the transmission contacts **320**, the first ground contact **250** and the second ground contact **260**, all of the parts mounted on the second substrate may be positioned in the inner space **330a**. The counterpart connector may be inserted into the inner space **330a**. In this case, a part of the counterpart connector may be inserted into the inner space **330a**, and a part of the substrate

connector **300** according to the second example may be inserted into the inner space of the counterpart connector. The ground housing **330** may be disposed to surround all sides with respect to the inner space **330a**.

(149) The insulation unit **340** supports the RF contacts **310**. The RF contacts **310**, the transmission contacts **320**, the first ground contact **250**, and the second ground contact **260** may be coupled to the insulation unit **340**. In the insulation unit **340**, the RF contacts **310**, the transmission contacts **320**, the first ground contact **250** and the second ground contact **260** may be coupled to the grounding housing **330** so as to be positioned in the inner space **330a**.

(150) The first ground contact **350** may implement a shielding function for the first RF contact **311** together with the ground housing **330**. The first ground contact **350** may be disposed between the first RF contact **311** and the transmission contact **320** on the basis of the first axial direction (X-axis direction). When the counterpart connector is inserted into the inner space **330a**, the first ground contact **350** may be connected to a ground contact of the counterpart connector.

(151) The first ground contact **350** may include a 1-1 ground contact **351** and a 1-2 ground contact **352**.

(152) The 1-1 ground contact **351** is disposed between the first RF contact **311** and the first transmission contacts **321** on the basis of the first axial direction (X-axis direction). Accordingly, the 1-1 ground contact **351** may shield between the first RF contact **311** and the first transmission contact **321**. The 1-1 ground contact **351** may include a 1-1 ground inspection plane **3511** for contacting the inspection instrument, a 1-1 ground mounting member **3512** for mounting on the second substrate, a 1-1 ground connecting member **3513** for connecting to a ground contact of the counterpart connector. The 1-1 ground contact **351** may include a plurality of 1-1 ground mounting members **3512**. The 1-1 ground contact **351** may further include a 1-1 ground inspection member **3514** for contacting the inspection instrument. Since the 1-1 ground inspection plane **3511**, the 1-1 ground mounting member **3512**, the 1-1 ground connecting member **3513** and the 1-1 ground inspection member **3514** may be implemented to substantially coincide with the 1-1 ground inspection plane **2511**, the 1-1 ground mounting member **2512**, and the 1-1 ground connecting member **2513** and the 1-1 ground inspection member **2514** in the substrate connector **200** according to the first example as described above, respectively, the detailed descriptions thereof will be omitted.

(153) The 1-2 ground contact **352** is disposed between the first RF contact **311** and the second transmission contact **322** on the basis of the first axial direction (X-axis direction). Accordingly, the 1-2 ground contact **352** may shield between the first RF contact **311** and the second transmission contact **322**. The 1-2 ground contact **352** may include a 1-2 ground inspection plane **3521** for contacting the inspection instrument, a 1-2 ground mounting member **3522** for mounting on the second substrate, and a 1-2 first ground connecting member **3523** for connecting to a ground contact of the counterpart connector. The 1-2 ground contact **352** may include a plurality of 1-2 ground mounting members **3522**. The 1-2 ground contact **352** may further include a 1-2 ground inspection member **3524** for contacting the inspection instrument. Since the 1-2 ground inspection plane **3521**, the 1-2 ground mounting member **3522**, the 1-2 ground connecting member **3523** and the 1-2 ground inspection member **3524** may be implemented to substantially coincide with the 1-2 ground inspection plane **2521**, the 1-2 ground mounting member **2522**, the 1-2 ground connecting member **2523** and the 1-2 ground inspection members **2524** in the substrate connector **200** according to the first example as described above, respectively, the detailed descriptions thereof will be omitted.

(154) The second ground contact **360** may implement a shielding function for the second RF contact **312** together with the ground housing **330**. The second ground contact **360** may be disposed between the second RF contact **312** and the transmission contacts **320** on the basis of the first axial direction (X-axis direction). When the counterpart connector is inserted into the inner space **330a**, the second ground contact **360** may be connected to a ground contact of the counterpart connector.

(155) The second ground contact **360** may include a 2-1 ground contact **361** and a 2-2 ground contact **362**.

(156) The 2-1 ground contact **361** is disposed between the second RF contact **312** and the second transmission contacts **322** on the basis of the first axial direction (X-axis direction). Accordingly, the 2-1 ground contact **361** may shield between the second RF contact **312** and the second transmission contact **322**. The 2-1 ground contact **361** may include a 2-1 ground inspection plane **3611** for contacting the inspection instrument, a 2-1 ground mounting member **3612** for mounting on the second substrate, and a 2-1 ground connecting member **3613** for connecting to a ground contact of the counterpart connector. The 2-1 ground contact **361** may include a plurality of 2-1 ground mounting members **3612**. The 2-1 ground contact **361** may further include a 2-1 ground inspection member **3614** for contacting the inspection instrument. Since the 2-1 ground inspection plane **3611**, the 2-1 ground mounting member **3612**, the 2-1 ground connecting member **3613** and the 2-1 ground inspection member **3614** may be implemented to substantially coincide with the 2-1 ground inspection plane **2611**, the 2-1 ground mounting member **2612**, the 2-1 ground connecting member **2613** and the 2-1 ground inspection member **2614** in the substrate connector according to the first example as described above, respectively, the detailed descriptions thereof will be omitted.

(157) The second 2-2 ground contact **362** is disposed between the second RF contact **312** and the first transmission contacts **321** on the basis of the first axial direction (X-axis direction). Accordingly, the 2-2 ground contact **362** may shield between the second RF contact **312** and the first transmission contact **321**. The 2-2 ground contact **362** may include a 2-2 ground inspection plane **3621** for contacting the inspection instrument, a 2-2 ground mounting member **3622** for mounting on the second substrate, and a 2-2 ground connecting member **3623** for connecting to a ground contact of the counterpart connector. The 2-2 ground contact **362** may include a plurality of the 2-2 ground mounting members **3622**. The 2-2 ground contact **362** may further include a 2-2 ground inspection member **3624** for contacting the inspection instrument. Since the 2-2 ground inspection plane **3621**, the 2-2 ground mounting member **3622**, the 2-2 ground connecting member **3623** and the 2-2 ground inspection member **3624** may be implemented to substantially coincide with the 2-2 ground inspection plane **2621**, the 2-2 ground mounting member **2622**, the 2-2 ground connecting member **2623** and the 2-2 ground inspection member **2624** in the substrate connector **200** according to the first example as described above, respectively, the detailed descriptions thereof will be omitted.

(158) In the substrate connector **300** according to the second example, as illustrated in FIG. **20**, the 1-1 ground inspection plane **3511**, the first transmission inspection planes **3212**, the 2-2 ground inspection plane **3621** and the second RF inspection plane **3122** may be disposed on the first row **R1**. The first RF inspection plane **3112**, the 1-2 ground inspection plane **3512**, the second transmission inspection planes **3222** and the 2-1 ground inspection plane **3611** may be disposed on the second row **R2**. In the substrate connector **300** according to the second example, the inspection planes **3112**, **3122**, **3212**, **3222**, **3511**, **3521**, **3611**, **3621** may be disposed as illustrated in FIG. **6**, **9** or **14**.

(159) Referring to FIGS. **16** to **21**, in the substrate connector **300** according to the second example, the ground housing **330** may be implemented as follows.

(160) The ground housing **330** may include a ground side wall **331**, a ground upper wall **332** and a ground lower wall **333**.

(161) The ground side wall **331** faces the insulation unit **240**. The ground side wall **331** may be disposed to face the inner space **330a**. The ground side wall **331** may be disposed to surround all sides of the inner space **330a** as a reference.

(162) The ground side wall **331** may be connected to a grounding housing of a counterpart connector inserted into the inner space **330a**. For example, as illustrated in FIG. **16**, the ground side wall **331** may be connected to a ground inner wall **231** of the ground housing **230** of the substrate connector **200** according to the first example. In this way, the substrate connector **300** according to

the second example may further strengthen the shielding function through the connection between the ground housing **330** and the ground housing of the counterpart connector. In addition, the substrate connector **300** according to the second example may reduce electrical adverse effects such as crosstalk or the like that may be generated by capacitance or induction between adjacent terminals through the connection between the ground housing **330** and the ground housing of the counterpart connector. In this case, since the substrate connector **300** according to the second example may secure a path through which electromagnetic waves are introduced to at least one ground of the second substrate and the first substrate, the EMI shielding performance may be further strengthened.

(163) The ground upper wall **332** is coupled to the ground side wall **331**. The ground upper wall **332** may be coupled to one end of the ground side wall **331**. The ground upper wall **332** may protrude from the ground side wall **331** toward the inner space **330a**. The ground upper wall **332** may be connected to a ground housing of a counterpart connector inserted into the inner space **330a**. Accordingly, in the substrate connector **300** according to the second example, since the ground upper wall **332** and the ground side wall **331** are connected to the ground housing of the counterpart connector, the shielding function may be further strengthened by increasing the contact area between the ground housing **330** and the ground housing of the counterpart connector. For example, as illustrated in FIG. **16**, the ground upper wall **332** may be connected to the ground bottom **234** of the ground housing **230** of the substrate connector **200** according to the first example.

(164) The ground lower wall **333** is coupled to the ground side wall **331**. The ground lower wall **333** may be coupled to the other end of the ground side wall **331**. The ground lower wall **333** may protrude from the ground side wall **331** to the opposite side of the inner space **330a**. The ground lower wall **333** may be disposed to surround all sides with respect to the ground side wall **331**. The ground lower wall **333** and the ground side wall **331** may be implemented as a shielding wall surrounding the side of the inner space **330a**. The first RF contact **311** and the second RF contact **312** may be positioned in the inner space **330a** surrounded by the shielding wall. Accordingly, the ground housing **330** may implement a shielding function for the RF contacts **310** using a shielding wall. Therefore, the substrate connector **300** according to the second example may contribute to further improving EMI shielding performance and EMC performance by using the shielding wall. The lower ground wall **333** may be grounded by being mounted on the second substrate. In this case, the ground housing **330** may be grounded through the ground lower wall **333**.

(165) The ground lower wall **333** and the ground upper wall **332** may be formed in a plate shape disposed in the horizontal direction, and the ground side wall **331** may be formed in a plate shape disposed in the vertical direction. The ground lower wall **333**, the ground upper wall **332** and the ground side wall **331** may be integrally formed.

(166) Herein, the ground housing **330** may implement a shielding function for the first RF contact **311** together with the first ground contact **350**. The ground housing **330** may implement a shielding function for the second RF contact **312** together with the second ground contact **360**.

(167) In this case, as illustrated in FIG. **21**, the ground housing **330** may include a first shielding wall **330b**, a second shielding wall **330c**, a third shielding wall **330d** and a fourth shielding wall **330e**. The first ground loop **350** may be implemented for the first ground contact **350**, the first shielding wall **330b**, the third shielding wall **330d** and the fourth shielding wall **330e**. The second ground loop **360** may be implemented for the second ground contact **360**, the second shielding wall **330c**, the third shielding wall **330d** and the fourth shielding wall **330e**. Since the first shielding wall **330b**, the second shielding wall **330c**, the third shielding wall **330d**, the fourth shielding wall **330e**, the first ground loop **350a** and the second ground **360a** are implemented to substantially coincide with the first shielding wall **230b**, the second shielding wall **230c**, the third shielding wall **230d**, the fourth shielding wall **240d**, the first ground loop **250a** and the second ground loop **260a** in the substrate connector **200** according to the first example as described above, respectively, the detailed

descriptions thereof will be omitted.

(168) Referring to FIGS. 2 to 21, in the substrate connector 300 according to the second example, the insulation unit 340 may include a soldering inspection window 341 which is used to inspect a state of being mounted on the second board. Since the soldering inspection window 341 is implemented to substantially coincide with the soldering inspection window 244 in the substrate connector 200 according to the first example, the detailed description thereof will be omitted.

(169) The present disclosure described above is not limited to the above-described exemplary embodiments and the accompanying drawings, and it will be apparent to those of ordinary skill in the art to which the present disclosure pertains that various substitutions, modifications and changes are possible within the scope that does not deviate from the technical spirit of the present disclosure.

Claims

1. A substrate connector, comprising: a plurality of radio frequency (RF) contacts for RF signal transmission; an insulation unit for supporting the RF contacts; a plurality of transmission contacts coupled to the insulation unit between a first RF contact and a second RF contact such that the first RF contact of the RF contacts and the second RF contact of the RF contacts are spaced apart from each other in a first axial direction; a ground housing to which the insulation unit is coupled; a first ground contact which is coupled to the insulation unit and shields the gap between the first RF contact and the transmission contacts on the basis of the first axial direction; and a second ground contact which is coupled to the insulation unit and shields the gap between the second RF contact and the transmission contacts on the basis of the first axial direction, wherein the first RF contact includes a first RF inspection plane with which an inspection instrument is to come in contact, wherein the second RF contact includes a second inspection plane with which the inspection instrument is to come in contact, and wherein the first RF inspection plane and the second RF inspection plane are arranged on a plane having the same height.
2. The substrate connector of claim 1, wherein when at least one of the first RF inspection plane and the second RF inspection plane has a width of 1 on the basis of the first axial direction, the substrate connector is formed in a plane having a length of 0.5 or more on the basis of the second axial direction which is perpendicular to the first axial direction.
3. The substrate connector of claim 1, wherein when at least one of the first RF inspection plane and the second RF inspection plane has a width of 1 on the basis of the first axial direction, the substrate connector is formed in a plane having a length of 1.5 or less on the basis of the second axial direction which is perpendicular to the first axial direction.
4. The substrate connector of claim 1, wherein first transmission contacts of the transmission contacts and second transmission contacts of the transmission contacts are spaced apart from each other along a second axial direction which is perpendicular to the first axial direction, wherein each of the first transmission contacts includes a first transmission inspection plane with which the inspection instrument is to come in contact, wherein each of the second transmission contacts includes a second transmission inspection plane with which the inspection instrument is to come in contact, and wherein the first transmission inspection planes and the second transmission inspection planes are arranged on a plane having the same height.
5. The substrate connector of claim 4, wherein the first RF inspection plane, the second RF inspection plane, the first transmission inspection planes and the second transmission inspection planes are arranged on a plane having the same height.
6. The substrate connector of claim 4, wherein the first RF inspection plane is disposed at a higher or lower position than the first transmission inspection planes, and wherein the second RF inspection plane is disposed at a higher or lower position than the second transmission inspection planes.

7. The substrate connector of claim 4, wherein the first RF inspection plane and the first transmission inspection planes are disposed on a first row which is parallel to the first axial direction, wherein the second RF inspection plane and the second transmission inspection planes are disposed on a second row which is parallel to the first axial direction, and wherein the first row and the second row are spaced apart from each other along the second axial direction.
8. The substrate connector of claim 4, wherein the first transmission inspection planes are disposed on a first row which is parallel to the first axial direction, wherein the second transmission inspection planes are disposed on a second row which is parallel to the first axial direction, wherein the first RF inspection plane and the second RF inspection plane are disposed on a third row which is parallel to the first axial direction, and wherein the first row, the second row and the third row are spaced apart from each other along the second axial direction.
9. The substrate connector of claim 4, wherein the first ground contact includes a 1-1 ground contact and a 1-2 ground contact which are disposed to be spaced apart from each other along the second axial direction, wherein the 1-1 ground contact includes a 1-1 ground inspection plane with which the inspection instrument is to come in contact, and wherein the 1-2 ground contact includes a 1-2 ground inspection plane with which the inspection instrument is to come in contact.
10. The substrate connector of claim 9, wherein the 1-1 ground inspection plane, the 1-2 ground inspection plane, the first RF inspection plane, the second RF inspection plane, the first transmission inspection planes and the second transmission inspection planes are disposed on a plane having the same height.
11. The substrate connector of claim 9, wherein the 1-1 ground inspection plane and the first transmission inspection planes are disposed on a first row which is parallel to the first axial direction, wherein the 1-2 ground inspection plane and the second transmission inspection planes are disposed on a second row which is parallel to the first axial direction, and wherein the first row and the second row are disposed to be spaced apart from each other along the second axial direction.
12. The substrate connector of claim 11, wherein the first RF inspection plane is disposed on the first row, and wherein the second RF inspection plane is disposed on the second row.
13. The substrate connector of claim 11, wherein the first RF inspection plane and the second RF inspection plane are disposed on a third row which is parallel to the first axial direction, and wherein the first row, the second row and the third row are disposed to be spaced apart from each other along the second axial direction.
14. The substrate connector of claim 1, wherein the first RF contact includes a first RF mounting member for mounting on a substrate and a first RF connecting member for connecting to an RF contact of a counterpart connector, and wherein the first RF inspection plane is formed on the first RF mounting member or the first RF connecting member.
15. The substrate connector of claim 1, wherein the first RF contact includes a first RF mounting member for mounting on a substrate, a first RF connecting member for connecting to an RF contact of a counterpart connector, and a first RF inspection member with which the inspection instrument is to come in contact.
16. The substrate connector of claim 1, wherein first transmission contacts of the transmission contacts and second transmission contacts of the transmission contacts are disposed to be spaced apart from each other along a second axial direction which is perpendicular to the first axial direction, wherein each of the first transmission contacts includes a first transfer mounting member for mounting on a substrate, a first transfer connecting member for connecting to a transmission contact of a counterpart connector, and a first transmission inspection plane with which the inspection instrument is to come in contact, and wherein the first transmission inspection plane is formed on the first transfer mounting member or the first transfer connecting member.
17. The substrate connector of claim 1, wherein first transmission contacts of the transmission contacts and second transmission contacts of the transmission contacts are disposed to be spaced

apart from each other along a second axial direction which is perpendicular to the first axial direction, and wherein each of the first transmission contacts includes a first transfer mounting member for mounting on a substrate, a first transfer connecting member for connecting to a transmission contact of a counterpart connector, a first transmission inspection plane with which the inspection instrument is to come in contact, and a first transmission inspection member on which the first transmission inspection plane is formed.

18. The substrate connector of claim 1, wherein the first ground contact includes a 1-1 ground contact and a 1-2 ground contact which are disposed to be spaced apart from each other along a second axial direction which is perpendicular to the first axial direction, wherein the 1-1 ground contact includes a 1-1 ground mounting member for mounting on a substrate, a 1-1 ground connecting member for connecting to a ground contact of a counterpart connector, and a 1-1 ground inspection plane with which the inspection instrument is to come in contact, and wherein the 1-1 ground inspection plane is formed on the 1-1 ground mounting member or the 1-1 ground connecting member.

19. The substrate connector of claim 4, wherein the first ground contact includes a 1-1 ground contact and a 1-2 ground contact which are disposed to be spaced apart from each other along a second axial direction which is perpendicular to the first axial direction, and wherein the 1-1 ground contact includes a 1-1 ground mounting member for mounting on a substrate, a 1-1 ground connecting member for connecting to a ground contact of a counterpart connector, a 1-1 ground inspection plane with which the inspection instrument is to come in contact, and a 1-1 ground inspection member on which the 1-1 ground inspection plane is formed.
